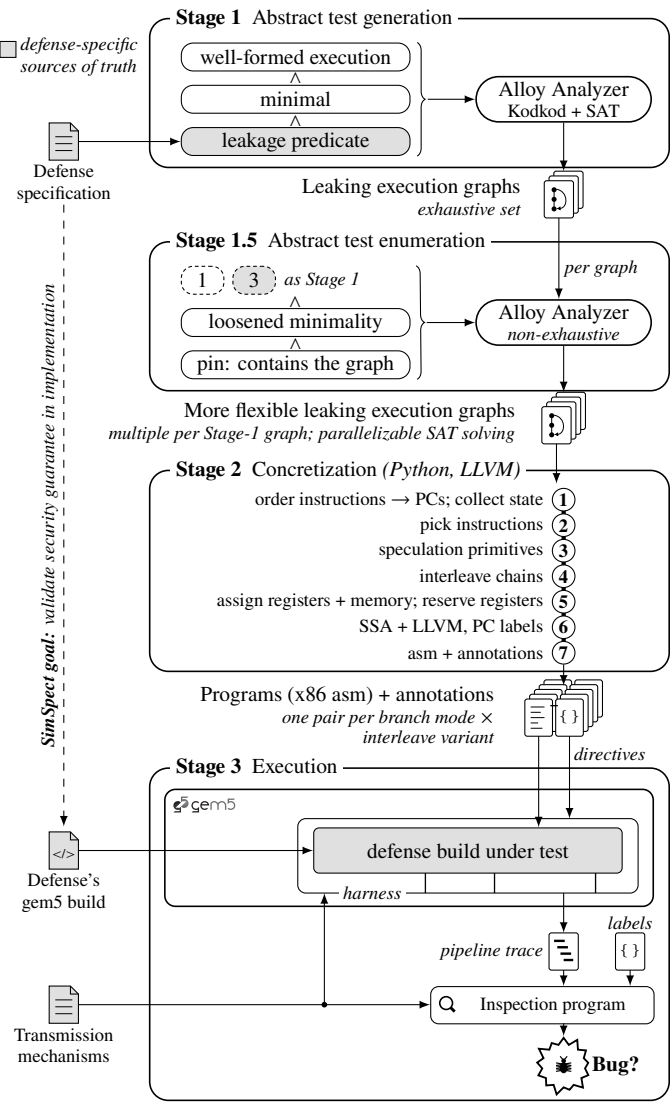

defense-specific sources of truth
Stage 1 Abstract test generation
well-formed execution
minimal
leakage predicate
Alloy Analyzer
Kodkod + SAT
Defense specification
Leaking execution graphs
exhaustive set
Stage 1.5 Abstract test enumeration
per graph
1
3
as Stage 1
loosened minimality
pin: contains the graph
Alloy Analyzer
non-exhaustive
More flexible leaking execution graphs
multiple per Stage-1 graph; parallelizable SAT solving
Stage 2 Concretization (Python, LLVM)
order instructions → PCs; collect state 1
pick instructions 2
speculation primitives 3
interleave chains 4
assign registers + memory; reserve registers 5
SSA + LLVM, PC labels 6
asm + annotations 7
SimSpect goal: validate security guarantee in implementation
Programs (x86 asm) + annotations
one pair per branch mode × interleave variant
directives
Stage 3 Execution
gem5
defense build under test
harness
Defense's gem5 build
labels
pipeline trace
Inspection program
Transmission mechanisms
Bug?